Math 142, F20 **Name:** Answer

Quiz 10

Instructions: **You must show all of your work to receive credits for correct answers.**

Problem 1. (10 points) Find the Maclaurin series and its radius of convergence for the following functions.

(a) $f(x) = x^2 e^x$

$$e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

for $|x| < +\infty$

$$\Rightarrow f(x) = x^2 e^x = x^2 \sum_{n=0}^{+\infty} \frac{x^n}{n!} = \sum_{n=0}^{+\infty} \frac{x^{n+2}}{n!} \quad \text{for } |x| < +\infty$$

$$\Rightarrow R = +\infty$$

(b) $f(x) = \sin(2x) - 2x$

$$\sin(x) = \sum_{n=0}^{+\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

for $|x| < +\infty$

$$\Rightarrow f(x) = \sin(2x) - 2x = \left[\sum_{n=0}^{+\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!}\right] - 2x$$

$$= \left[(2x) - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \frac{(2x)^7}{7!} + \cdots\right] - 2x$$

$$= -\frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \frac{(2x)^7}{7!} + \cdots$$

$$= \sum_{n=1}^{+\infty} \frac{(-1)^n (2x)^{2n+1}}{(2n+1)!} \quad \text{for } |2x| < +\infty$$

$$\Leftrightarrow |x| < +\infty$$

$$\Rightarrow R = +\infty$$

(c) $f(x) = \dfrac{x}{1+2x}$

$$\frac{1}{1-x} = \sum_{n=0}^{+\infty} x^n = 1 + x + x^2 + x^3 + \cdots$$

for $|x| < 1$

$$\Rightarrow f(x) = \frac{x}{1+2x} = x \cdot \left(\frac{1}{1-(-2x)}\right)$$

$$= x \cdot \sum_{n=0}^{+\infty} (-2x)^n$$

$$= \sum_{n=0}^{+\infty} (-2)^n x^{n+1}$$

for $|-2x| < 1$

$$\Rightarrow |x| < \frac{1}{2}$$

$$\Rightarrow R = \frac{1}{2}$$